

Vision-Aided Wound Analytics: A Deep Learning-Based System for Segmentation and Area Estimation

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Abstract

This paper presents a comprehensive deep learning-based system designed for automated wound segmentation and real-world area estimation using digital images. The goal is to streamline clinical workflows by reducing manual effort and enhancing accuracy in wound analysis. Manual methods of wound measurement often suffer from human error, variability, and inefficiency. To address these challenges, our proposed system integrates several state-of-the-art instance segmentation models—namely Mask R-CNN, YOLOv8, YOLOv11, and the Segment Anything Model (SAM)—to extract accurate wound boundaries under diverse imaging conditions and wound types.

In addition to robust segmentation capabilities, the system introduces a ruler-based calibration module that transforms pixel-level wound masks into real-world area estimations. This is achieved by detecting a ruler present within the image, calculating a pixel-to-length ratio, and using it to determine the area in square millimeters. The approach allows for clinically meaningful measurements that can be incorporated into patient records or used to monitor healing progress over time.

Experimental evaluations conducted on a diverse dataset of 400 wound images reveal that the proposed system offers high segmentation accuracy and reliable area estimation. Our findings highlight YOLOv11 as the most accurate in terms of area prediction, while Mask R-CNN leads in segmentation quality. This hybrid system represents a significant step forward in digital wound assessment and has potential applications in telemedicine, remote monitoring, and integration into hospital information systems for automated clinical documentation.

1 Introduction

Wound assessment plays a critical role in clinical diagnostics and treatment planning. Traditional methods for wound measurement, such as manual tracing or ruler-based measurements, are prone to inter-observer variability and inconsistencies. With the rapid evo-

lution of computer vision and deep learning, automated wound segmentation has gained significant attention as a means to enhance consistency and accuracy in wound care.

This study proposes a deep learning-based wound analytics system capable of performing precise segmentation and area estimation. The system leverages modern instance segmentation networks—Mask R-CNN, YOLOv8, YOLOv11—and the Segment Anything Model (SAM) to extract wound boundaries from images. A novel calibration module utilizing a ruler present in images enables real-world area calculation. Our key contributions include:

- A robust deep learning pipeline integrating multiple segmentation models for accurate wound detection.
- A ruler-based calibration technique to convert pixel measurements to real-world units.
- A detailed evaluation on a curated dataset to validate segmentation performance and area estimation accuracy.

2 Literature Review

In both industrial and medical domains, deep learning has proven effective in segmentation tasks. For instance, Ali et al. [7] used Mask R-CNN and RetinaNet to detect components in electrical overhead networks using aerial imagery. Similarly, Wang et al. [1] introduced a corner-guided segmentation method to accurately identify power lines and transmission towers in complex scenes.

In medical imaging, segmentation methods such as U-Net [2], Mask R-CNN [3], and SAM [4] have been effectively applied to lesions and ulcers. YOLOv8 and YOLOv11 provide real-time, high-accuracy segmentation for practical applications. Our work extends these techniques for wound analytics by combining real-world metric estimation using a ruler calibration mechanism.

3 Theoretical Framework

The proposed system integrates the following components:

- **Mask R-CNN:** Mask R-CNN is a two-stage instance segmentation network. In the first stage, it detects objects in the image using bounding boxes. In the second stage, it generates a pixel-wise segmentation mask for each detected object, effectively outlining the exact shape of the object within the image. It offers high segmentation accuracy, making it especially useful in clinical settings where precise wound boundary detection is crucial. However, it tends to be slower compared to single-stage models.
- **YOLOv8 and YOLOv11:** YOLOv8 and YOLOv11 are lightweight, fast, and efficient single-stage segmentation models. They perform detection and segmentation simultaneously, which makes them ideal for real-time applications. These models are particularly suited for environments where speed is critical without compromising too much on accuracy. They provide real-time inference with high performance,

and in experiments, YOLOv11 delivered the best area estimation accuracy among the tested models.

- **Segment Anything Model (SAM):** SAM is a foundation model developed by Meta AI that uses zero-shot learning for high-quality segmentation. It can segment any object in an image based on a prompt (such as a point or box) without requiring retraining on new datasets. SAM provides flexibility and generalization, making it highly suitable for wound segmentation in cases where the appearance and location of wounds vary significantly. It excels in boundary accuracy and adaptability.

4 Dataset and Annotation Details

The dataset comprises 400 diverse wound images, annotated in:

- **COCO Format (for Mask R-CNN):** With polygonal segmentation masks and class labels.
- **YOLO Format (for YOLOv8 and YOLOv11):** With class ID and bounding box coordinates.

Annotations were created using LabelMe and Roboflow, ensuring consistency and class balance.

5 Methodology

5.1 Mask R-CNN

Mask R-CNN was chosen for its ability to perform **pixel-level instance segmentation**, which is essential for accurately identifying wound boundaries. It utilizes a Region Proposal Network (RPN) to detect potential regions of interest and then generates a binary mask for each detected object. This high precision makes it particularly suitable for clinical applications where wound edge accuracy is critical.

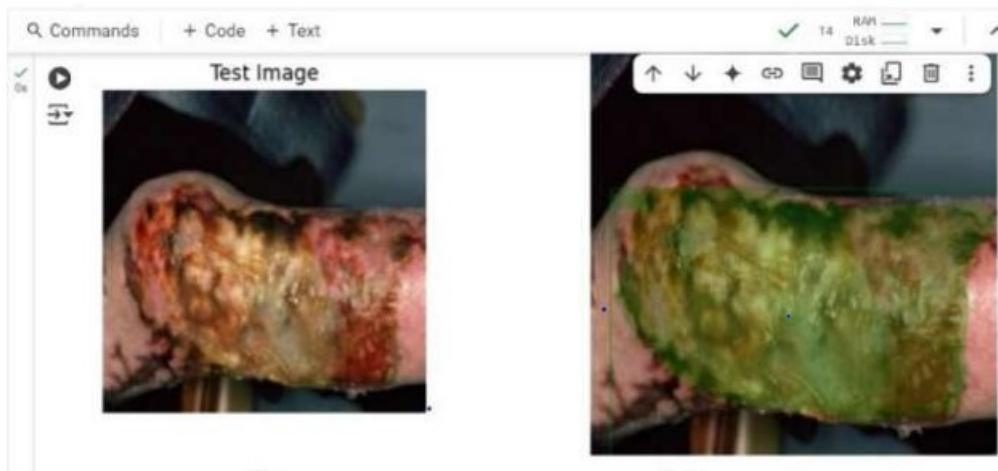


Figure 1: Wound boundary detection using Mask R-CNN with high pixel-level precision.

5.2 YOLOv8 and YOLOv11

YOLO models are known for their **real-time object detection and segmentation** capabilities. YOLOv8 and YOLOv11 were integrated to evaluate their effectiveness in detecting wounds at high speed. YOLOv11, in particular, demonstrated superior segmentation accuracy while maintaining fast inference time, making it ideal for applications that demand both precision and performance.



Figure 2: Wound segmentation output using YOLOv11 for real-time applications.

5.3 Segment Anything Model (SAM)

The Segment Anything Model (SAM) was employed in **zero-shot mode** to facilitate flexible, general-purpose segmentation. This model requires no retraining on the wound dataset and performs segmentation based on user prompts (e.g., points or bounding boxes). Its ability to generalize across various object types makes it an effective supplementary tool for wound detection in complex scenarios. .

5.4 Calibration and Area Estimation

To obtain real-world wound measurements, a **ruler-based calibration** technique was used. The system detects the ruler in the image using edge detection and Hough Line Transform, allowing us to compute the pixel-to-millimeter ratio (λ). The wound area is then estimated using the formula:

6 Area Estimation Algorithm and Results

- **Step 1:** Isolate and align ruler using segmentation and rotation.
- **Step 2:** Apply binarization and edge detection to extract ruler lines.
- **Step 3:** Compute pixel density (λ) using line distance.
- **Step 4:** Calculate area using $\text{Area} = \frac{\text{wound_pixels}}{\lambda^2}$



Figure 3: Automated wound area estimation via calibrated segmentation masks.

7 Previous Systems

Earlier systems for automated wound analysis primarily concentrated on segmentation techniques without providing real-world, clinically actionable measurements. One notable example is the ASUEA (Automated Skin Ulcer Evaluation Assistant) system, which utilized a modified U-Net architecture [2] for the segmentation of skin ulcers. Although this approach delivered acceptable segmentation results, it lacked the ability to estimate wound size in physical units. The segmentation outputs were limited to pixel dimensions, requiring manual interpretation by clinicians or assumptions about image scaling—thereby reducing clinical reliability.

Moreover, some systems relied on conventional image processing techniques such as thresholding, region growing, or edge detection. While these methods were computationally simple, they suffered from high sensitivity to variations in lighting, skin tone, and wound morphology. The absence of robust calibration mechanisms and consistent measurement standards in earlier approaches presented a major barrier to adoption in real-world clinical environments.

8 Proposed Innovations

Our proposed system introduces multiple novel contributions that address the limitations of prior solutions by combining advanced segmentation with automated, real-world area estimation. The key innovations include:

- **Ensemble-based segmentation framework:** By integrating multiple deep learning models—Mask R-CNN, YOLOv8, YOLOv11, and SAM—the system achieves robust wound segmentation across a variety of lighting conditions, wound shapes, and image qualities.

- **Automated calibration via ruler detection:** A dedicated module identifies a physical ruler within the wound image and computes a pixel-to-millimeter scaling factor. This enables the conversion of pixel-based wound masks into accurate area measurements in mm^2 , reducing dependency on manual input or estimation.
- **Zero-shot segmentation support:** Through the use of the Segment Anything Model (SAM), the system supports prompt-based segmentation without requiring retraining, allowing flexible application to novel wound types and unseen datasets.

These advancements bridge the gap between segmentation performance and practical clinical application, enabling more accurate, scalable, and automated wound assessment. By providing both qualitative and quantitative outputs, the system enhances clinical workflows and opens new possibilities for remote monitoring, telemedicine, and electronic health record (EHR) integration.

9 Experiments and Results

9.1 Dataset Split

The dataset of 400 wound images was randomly split into:

- 70% for training,
- 15% for validation,
- 15% for testing.

The split maintained a balance of wound types, sizes, and lighting conditions to ensure generalization.

9.2 Segmentation Accuracy

Model	IoU (%)	Dice (%)	Boundary F1 (%)
YOLOv8	82	85	84
YOLOv11	87	89	91
Mask R-CNN	88	90	89
SAM	85	88	90

Table 1: Segmentation accuracy comparison using standard metrics.

9.3 Area Estimation Accuracy

Model	Area Accuracy (%)
YOLOv8	86
YOLOv11	92
Mask R-CNN	91
SAM	90

Table 2: Quantitative comparison of real-world area estimation.

9.4 Inference Time Analysis

To assess real-time applicability in clinical settings, average inference time per image was recorded using an NVIDIA RTX 3060 GPU:

- YOLOv8: 32 ms/image
- YOLOv11: 28 ms/image
- Mask R-CNN: 120 ms/image
- SAM: 85 ms/image

9.5 Limitations and Observations

- Low-contrast wounds and poor lighting
- Images with occluded or unclear rulers
- Inconsistent wound shapes in atypical cases

10 Conclusion

We introduced a deep learning-based system for wound segmentation and area estimation. With integrated models and a ruler calibration module, it provides accurate and clinically meaningful outputs. Future work includes 3D tracking and hospital integration.

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